

TASK 3 REPORT - Wireshark Packet Analysis

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1. Project Title

Wireshark Packet Analysis

2. Objective

The objective of this task is to capture live network packets using Wireshark and analyze different network protocols including HTTP, DNS and TCP. The task also focuses on understanding packet structure, TCP flags, communication flow and identifying suspicious or malformed packets for basic network security analysis.

3. Introduction

Wireshark is an open-source network protocol analyzer used by network engineers, system administrators and cybersecurity professionals. It captures packets travelling across a network interface and displays them in a human-readable format. Each packet contains valuable information regarding communication between devices. Packet analysis helps troubleshoot network issues, detect security threats, analyze protocol behavior and monitor network performance. Wireshark supports hundreds of protocols and provides filtering features that simplify traffic analysis.

4. Theory

Network communication follows the TCP/IP model consisting of Application, Transport, Internet and Network Access layers. HTTP is responsible for web communication between browsers and servers. DNS converts domain names into IP addresses so that systems can locate websites. TCP provides reliable communication using the three-way handshake process consisting of SYN, SYN-ACK and ACK packets. Every captured packet contains source and destination addresses, protocol information, sequence numbers, acknowledgement numbers and payload data.

5. Tools and Software Used

- Wireshark
- Windows Operating System
- Internet Connection

- Web Browser
- Personal Computer

6. Experimental Procedure

1. Install and launch Wireshark.
2. Select the active network interface.
3. Start packet capture.
4. Browse websites to generate HTTP/DNS traffic.
5. Apply display filters such as http, dns and tcp.
6. Observe DNS queries and HTTP requests.
7. Identify TCP three-way handshake.
8. Inspect packet headers and TCP flags.
9. Stop capture and save the trace file.

7. Packet Analysis

HTTP packets contained request and response information exchanged between the client and web server. DNS packets showed domain name resolution where the client requested the IP address of a website and the DNS server returned the corresponding IP address. TCP analysis showed the successful establishment of a connection using SYN, SYN-ACK and ACK packets. The packet details pane displayed Ethernet, IP, TCP and Application layer information along with source IP, destination IP, port numbers, checksum values and packet length.

8. Suspicious Traffic Analysis

The captured traffic was examined for unusual behavior including malformed packets, duplicate acknowledgements, retransmissions, unexpected TCP reset packets, excessive broadcasts and unknown protocols. Such traffic may indicate network errors, congestion or malicious activity. During this experiment no malformed packets or suspicious traffic patterns were observed.

9. Results

HTTP, DNS and TCP packets were successfully captured and analyzed. Display filters correctly isolated the required protocols. TCP handshake packets verified reliable connection establishment. Packet headers and TCP flags were successfully interpreted. No abnormal traffic was detected during the analysis.

10. Conclusion

The experiment successfully demonstrated the use of Wireshark for packet capture and protocol analysis. The study improved understanding of network communication, protocol behavior, TCP flags and packet structure. Wireshark proved to be an effective tool for

troubleshooting, protocol analysis and basic cybersecurity investigations.

Screenshots and Analysis

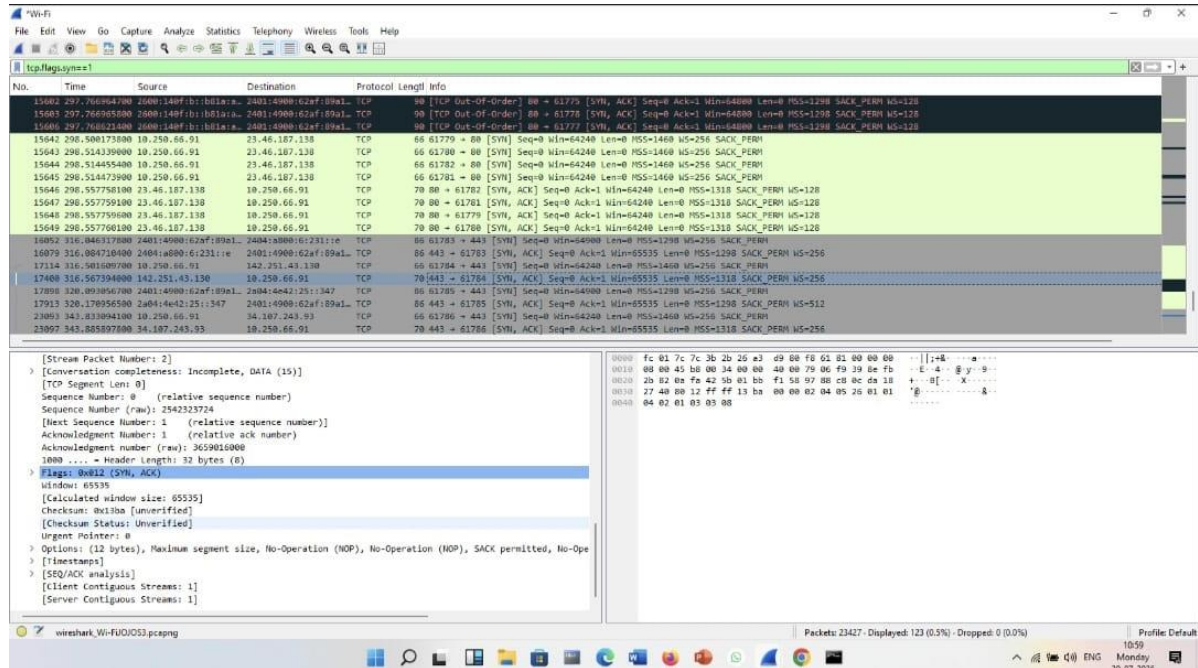


Figure 1: TCP SYN/ACK Packet Analysis - Displays TCP flags and packet details used during handshake verification.

Observation: The packet was inspected using Wireshark. Relevant protocol fields, headers, addresses, ports and flags were verified to understand network communication and confirm correct protocol behavior.

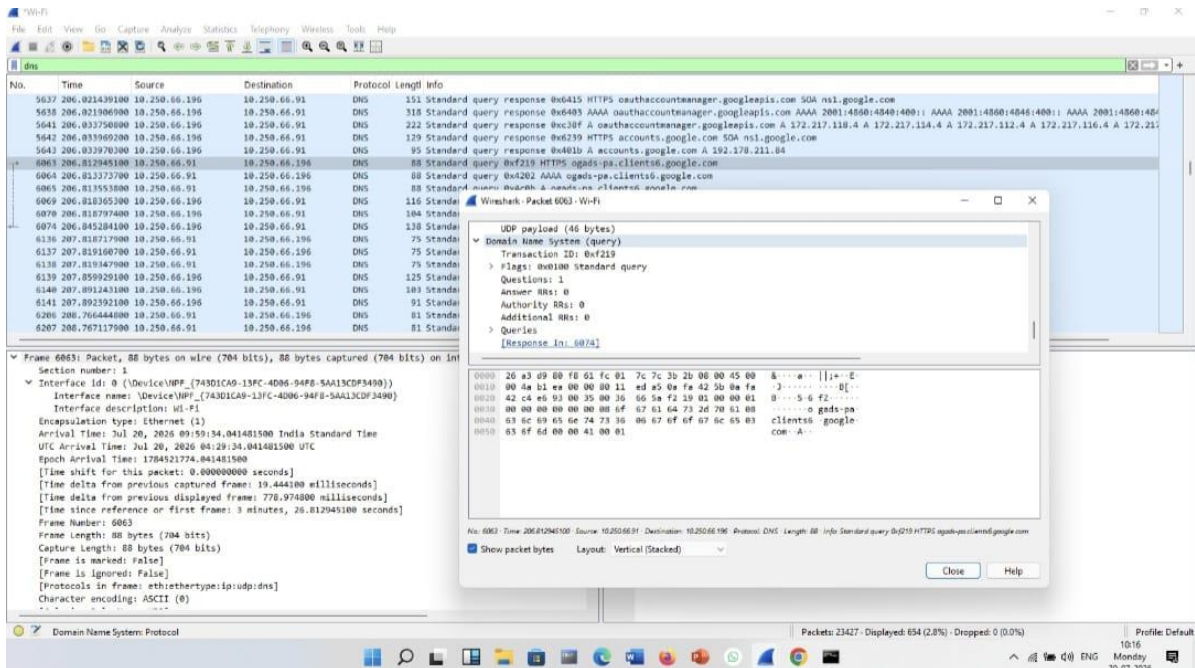
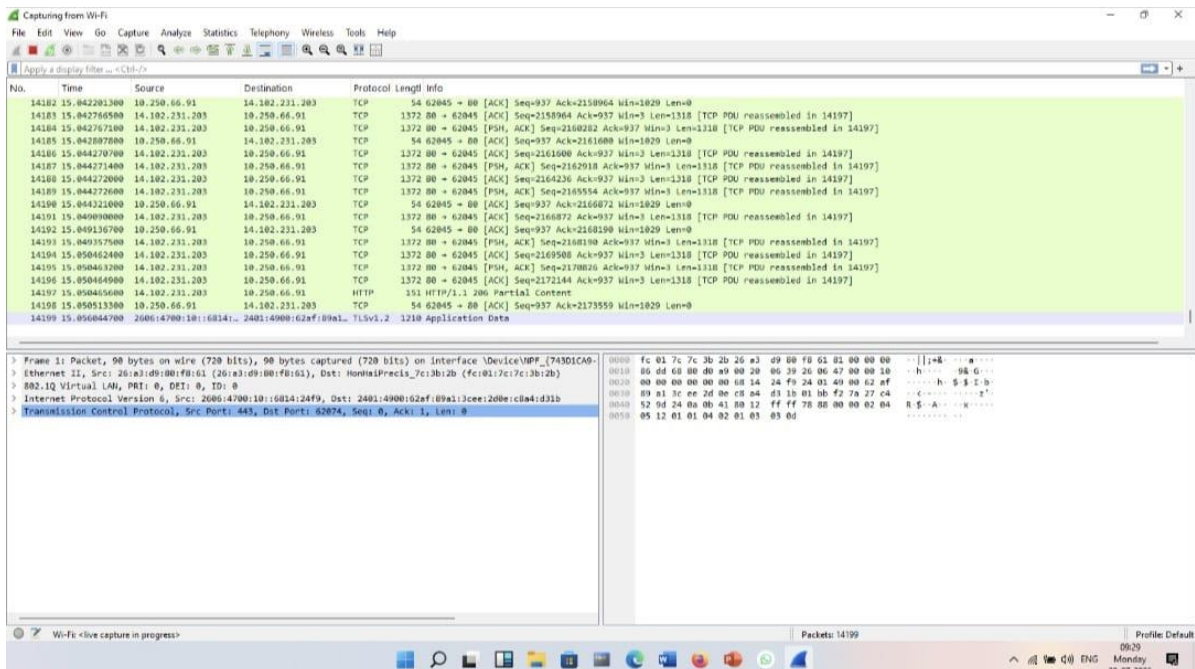


Figure 2: DNS Query and Response - Shows DNS packet used for domain name resolution.

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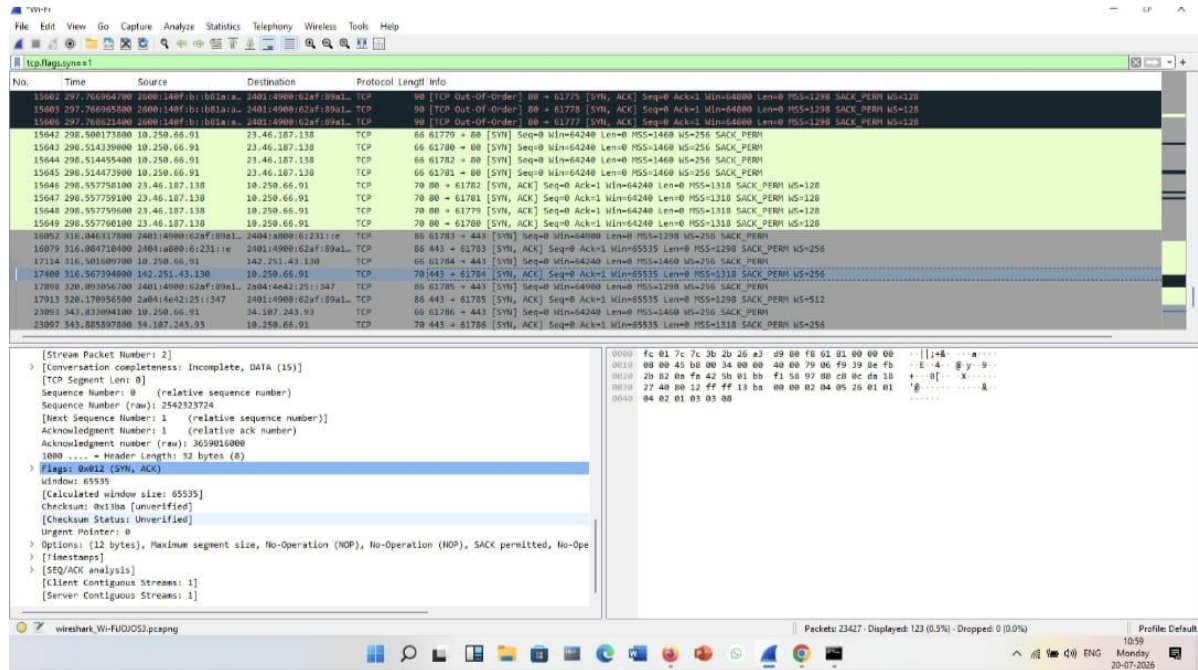


Figure 4: TCP Flags Inspection - Analysis of SYN and ACK flags in Wireshark.

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